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Neuro-cognitive performance is enhanced during short periods of microgravity

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HIGHLIGHTS

- Neurocognitive function was found to be improved in microgravity.
- No effect of anti-nausea medication was observed.
- No effect of flight experience was observed.
- Further research is necessary addressing the effects of permanent microgravity.

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ABSTRACT

There is increasing interest in the effects of microgravity on cognitive processing, particularly as it relates to the potential for human space travel. While findings to date are quite inconsistent, studies reporting a decrement in cognitive performance have generally not been able to distinguish between the direct influence of microgravity, and any associated influence of stress. Furthermore, the currently available findings are primarily based on behavioral observations, and there is a need to better understand the underlying neurophysiological responses.
The current study aimed to determine the effects of microgravity on neurophysiological processing during a mental arithmetic task (executive function). During the normal- and microgravity phases of a parabolic flight, four levels of a mental arithmetic task were presented on a touchscreen tablet. The latency between the appearance of the problem and the participants' response was identified as reaction time. In addition visual evoked potentials N1 and P2 were determined using an active EEG system and analyzed using source localization algorithms.
Results showed an increase in reaction time with increasing levels of task difficulty. During the most complex levels, reaction time was significantly reduced during microgravity. This observation was independent of previous parabolic flight experience as well as the use of anti-motion-sickness medication. P2 amplitude decrease during microgravity was concomitant to a related involvement of the superior frontal and medial frontal gyrus.
It is concluded that cortical processes are enhanced during microgravity, and that previously reported impairments in cognitive performance are likely attributable to increased stress rather than weightlessness itself.
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1. Introduction

In recent decades, several studies have evaluated cognitive processes in microgravity, including during space travel and during experimental parabolic flights. While it is often assumed that cognitive function is impaired in microgravity, the findings to date have been inconsistent, with most studies reporting no direct effect of microgravity on cognitive function [1,2]. Upon closer examination of previous experiments, it is apparent that a reduction in cognitive function is most likely to occur: (1) during the so called 'transition-phases' of space flight, i.e. observations made in the periods soon after arriving in space or just prior to leaving space [1,3,4], or (2) during the 0G (weightlessness) phase of parabolic flights, compared with ground observations [5–12]. It is likely that measurements during these periods are subject to a mixture of possible influences, namely stress and microgravity, and it is nearly impossible to distinguish between the two. In an attempt to control for the stress-related effects that are associated with parabolic flight [30], a primary aim of the current experiment was to compare cognitive

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function and any underlying neural correlates, identified by event related potentials (ERP), between 1G- and 0G-conditions during parabolic flight. As these conditions are separated by <30s during the same parabola, it is likely that any differences can be attributed directly to the changes in G-load, i.e. the direct effect of microgravity.

To further understand the impact of weightlessness, it is important to not only to describe the behavioral parameters of cognitive performance (e.g. reaction time), but also to investigate the underlying neurophysiological effects. As the use of brain imaging techniques is limited in extreme environments, there has been an increased emphasis on the development and use of valid and reliable portable assessment systems. Electroencephalography (EEG), which allows for the assessment of ERPs with a high temporal resolution, is well suited for the study of brain activity in extreme environments. First studies on microgravity related ERPs already provide us with an idea that central processes in (simulated) weightlessness are altered [37–39].

If the stress related effects of parabolic flight are able to be controlled, it might be expected that brain function and cognitive performance should increase in microgravity, and this may be a function of improved oxygen delivery to the brain during weightlessness [15]. As such, we hypothesized that cognitive function would be enhanced during the microgravity phase of a parabolic flight, as indicated by improved performance during a simple to complex reaction time task, which we anticipated would also be mirrored by changes in neurocognitive markers.

2. Methods

2.1. Participants and procedure

European Space Agency (ESA) and German Space Agency (DLR) parabolic flights take place from Bordeaux International Airport (F), aboard the A300 ZeroG. A parabolic flight maneuver is characterized by gravitational changes from 1G to 1.8G, to 0G (weightlessness, approx. 22 s), and back to 1.8G, and finally to 1G. One campaign consists of three to four flight days, and on each flight day 30 experimental parabolas are completed. During eight parabolic flight campaigns scheduled between November 2010 and November 2013, data from 43 participants were recorded (female: n = 13, age 35.0 +/− 8.7; male: n = 31, age 39.9 +/− 9.2 years). All participants, either staff members or students of the involved universities, underwent a pre-campaign clinical examination and provided informed consent. Prior, the experimental design of the study was approved by the Research Ethics Committee of the German Sport University, in accordance with the Declaration of Helsinki. During the flight, participants were seated in a relaxed position and were firmly and comfortably strapped to the floor in order to avoid artifacts caused by stabilization movements. To standardize the position of the tablet screen on which the arithmetic problem-solving task was presented, the tablet was adhered to the participants' non-dominant forearm. As described below, cognitive performance was assessed throughout the flight using an arithmetic problem solving task, and EEG signals were continuously collected throughout the flight. All participants were familiarized with the arithmetic task and the experimental protocol 24 h prior to take-off.

2.2. Cognitive task

The experimental cognitive task consisted of a mental arithmetic task (problem solving) with two numbers presented on the left and right side of a tablet screen. Participants were to respond by deciding which number was higher, and pressing the corresponding side (left or right) of the tablet screen. The task was divided into four levels with increasing complexity. During the first level participants had to decide between two simple numbers (e.g. 13 vs. 7). With each subsequent level, the difficulty of the task increased so that during the final level (level 4) participants has to decide between complex arithmetic equations (e.g. 8 ∗ (12 − 4) vs. 18 − 2 ∗ 5). Participants were instructed to respond and make their selection as quickly and as accurately as possible. To avoid any external disturbances a flight operator supervised each participant. Appearance of each pair of numbers, as well as the corresponding reaction (touch), was triggered to the EEG signal.

Participants completed the task during the 1G phase and the 0G phase of four consecutive parabolas. Participants were randomly assigned to two subgroups: 16 participants (age 36.3 +/− 10.5 years) repeatedly performed the task at the easiest level (group EASY); and 26 participants (aged 38.5 +/− 8.4 years) performed the task with increasing levels of difficulty during each consecutive parabola (group DIFFICULT). To avoid any order effect, participants in each group were randomly assigned to commence the task during the 0G phase (i.e. 0G followed by 1G), or during the 1G phase (i.e. 1G followed by 0G).

In order to test for possible effects of flight experience and medication (Scopolamine), which is normally administered before parabolic flight in order to prevent motion sickness, this second group (DIFFICULT) was further subdivided into three groups. The first subgroup consisted of participants with no previous parabolic flight experience and without any medication (first time flyers — no medication, n = 10, age 36.2 +/− 7.0 years). The second subgroup also consisted of first time flyers, and participants in this group received Scopolamine, which was administered 60 min before take-off (first time flyers —with medication, n = 8, age 41.4 +/− 6.3 years). The third subgroup consisted of participants with previous parabolic flight experience, who had participated in a minimum of 100 parabolas, without any medication (frequent flyers, n = 8, age 39.6 +/− 11.8 years).

2.3. EEG data collection

An EEG actiCAP (Brain Products GmbH, Munich, Germany) with 32 Ag-AgCl electrodes, arranged in the international 10–20 system at positions Fp1, Fp2, F3, F4, F7, F8, Fz, FC1, FC2, FC5, FC6, C3, C4, Cz, CP1, CP2, CP5, CP6, P3, P4, P7, P8, Pz, PO9, PO10, T7, T8, TP9, TP10, O1, O2 and Oz, was mounted on the participant's head prior to take off. Each electrode was referenced to a reference electrode that was mounted in the triangle of FP1, FP2 and Fz. There was also a ground electrode. Electrodes were filled with Electro-Gel™ (Electro-Cap International, USA) for optimal signal transduction. During the flight, impedance was kept below ten kiloohm [kΩ] by refilling the electrodes with gel. The analog signal

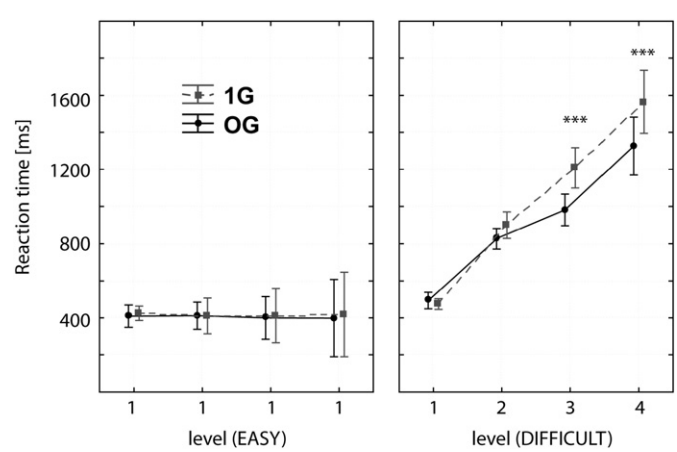

Fig. 1. Reaction time (RT) for the mental arithmetic task performed in normal gravity (dashed line) and microgravity (solid line). Group EASY (left) repeated the task at the easiest level, whereas the level of difficulty was progressively increased for group DIFFICULT (right). With increasing levels of difficulty reaction time increased. ***p < .001 between groups performed in 0G and 1G at levels 3 and 4.

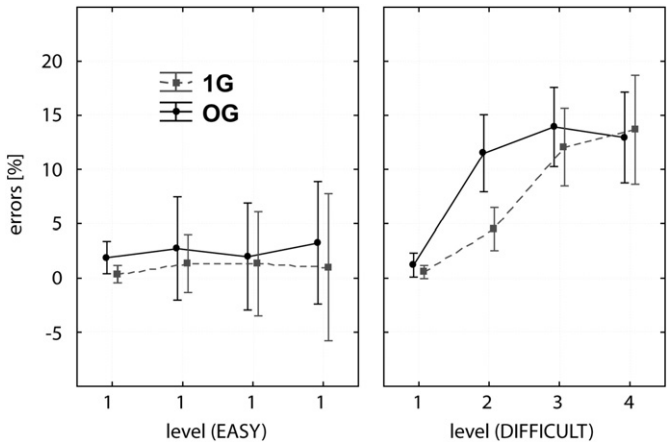

Fig. 2. Accuracy indicated as percentage of errors for the mental arithmetic task performed in normal gravity (dashed line) and microgravity (solid line). Group EASY (left) repeated the task at the easiest level, whereas the level of difficulty was progressively increased for group DIFFICULT (right). With increasing levels of difficulty the percentage of errors increased but no significant difference was found between 0G and 1G.

of the EEG was amplified, converted to digital signals and stored using a Brain Vision amplifier and RecView software (Brain Products GmbH, Munich, Germany). All recordings were monitored by a trained investigator.

2.4. EEG data analysis

EEG signals were analyzed offline using Brain Vision Analyzer 2.0 (Brain Products, Munich, Germany). After the application of low and high cut off filters, a frequency range between 5 and 30.0 Hz remained for analyses (time constant 0.3183099 s; 48 dB/octave). Any channels exceeding an impedance of 10 kΩ were excluded from further analysis. Following visual inspection, automatic artifact rejection was applied (gradient <50 μV; max/min amplitude −200 to 200 μV; lowest allowed activity in intervals 0.5 μV). From the 43 participants, EEG data of 34 participants was eligible for further ERP analysis.

Event related potentials (ERP) usually consist of several components that are assumed to be real-time correlates of specific cognitive processes. The relatively early components mainly reflect automatic processes such as stimulus detection, while the later components primarily reflect control processes.

As ERPs are time-locked to stimuli, we identified the appearance of the task (i.e. the display of numbers on the screen) as the relevant stimulus. Following segmentation based on stimulus onset (−100 to 800 ms), segmented data were baseline corrected (−100 to 0 ms) and averaged, using a minimum of 50 epochs for each participant for the 0G and a minimum of 50 epochs for each participant for the 1G condition over all electrodes. Signal to noise ratios (SNR) for each participant and each condition were calculated using the equation $P_{signal}/P_{noise.}$ Even in the worst case — the similarity between P_{signal} and P_{noise} was still below 2%.

As recorded scalp potentials show different volumes based on the localization of the reference, all averaged ERP waveforms were transformed into reference-free current source density (CSD) maps (order of splines: 4; maximal degree of legendre polynomials: 10; Lambda 1e-5). This allows for the generation of topographies with more sharply localized peaks than those of scalp potentials and eliminates volume-conducted activity from distant sources.

It was expected that the presentation of a visual task would elicit a visual evoked potential (VEP) which is characterized by a N1–P2 complex, where N1 is the first negative-going component at around 150 to 200 milliseconds post-stimulus and detectable in centro-frontal scalp areas. The visual N1 is a component evoked by any visual stimulus, nevertheless its amplitude might serve as an indicator of task complexity [16]. The visual P2 is a positive electrical potential that peaks between 150 and 275 ms post-stimulus in centro-frontal and centro-parietal regions and has been identified to be related to higher-order perceptual processing and to be influenced by the complexity of tasks [17]. Accordingly, amplitude (μV) as well as latency (ms) of these visual evoked potentials were exported for statistical analysis of N1 and P2.

2.5. Localization of EEG activity — sLORETA analysis

A fundamental limitation of traditional EEG recording procedures is the inverse solution. That is, there is a challenge in determining activated brain regions by extra cranial recordings. Standardized low resolution brain electromagnetic tomography (sLORETA) enables the spatial identification and analysis of brain cortical activity via traditional EEG recordings [18–21]. Extra cranial EEG activity is evoked by synchronized postsynaptic potentials (PSPs), which are located in clusters of pyramidal cells [22]. sLORETA allows for the determination of the three dimensional orientation of these highly synchronized PSPs. Cortical regions are created by allocating the raw LORETA values of individual voxels to their corresponding Broadmann areas (BA) or cerebral gyri on the basis of the coordinates of the digitized Talairach Atlas [23]. First, the coordinates of the electrode positions were applied to a digitized MRI version of the Talairach Atlas (McConnell Brain Imaging Centre, Montréal Neurological Institute, McGill University). These Talairach coordinates were then used to compute the sLORETA transformation matrix. Using the sLORETA transformation matrix, averaged ERP signals of each individual for the 0G and 1G condition were transformed to sLORETA files. This resulted in the corresponding 3D cortical distribution of the electrical neuronal generators for each subject. The computed sLORETA image displayed the cortical neuronal oscillators in 6430 voxels, with a spatial resolution of 5 mm.

2.6. Statistics

Statistical analyses were performed using STATISTICA version 7.1 (StatSoft, Tulsa, USA). Comparisons of reaction time, accuracy (percentage of errors) were performed using two-way repeated measures ANOVA with the within-group factors of “level” (1–4) and “gravity” (1G/0G), and the between-group factor complexity (EASY/DIFFICULT). Further analysis of the effects of experience or medication were calculated using the factors “gravity” (0G/1G), “level” (1–4, RT only) and “group” (frequent flyer, first time flyer – no medication, first time flyer – with medication). Analysis of N1 and P2 amplitude and latency during the EASY trial was performed using repeated measures ANOVA with the within-group factor of “gravity” (1G/0G).

Where appropriate, Fisher's least significance difference test (LSD) was used for post-hoc analyses. The level of significance was set at $p < 0.05$. Data are presented as mean +/− standard deviation (SD).

To localize differences in ERP activity between normal gravity and weightlessness, paired sampled t-test were computed on Log transformed data at each voxel. Statistical significance was assessed using a nonparametric randomization test [24]. To correct for multiple comparisons, a nonparametric single-threshold test was assessed defining a *critical threshold* ($t_{critical}$). Voxels with statistic values exceeding this threshold have their null hypothesis rejected. The *omnibus hypothesis* (that all the voxel hypotheses are true) is rejected if a voxel value exceeds the critical threshold for $p < .05$ defined by 5000 randomizations. Voxel-by-voxel t-values in Talairach space are displayed as statistical parametric maps (SPMs).

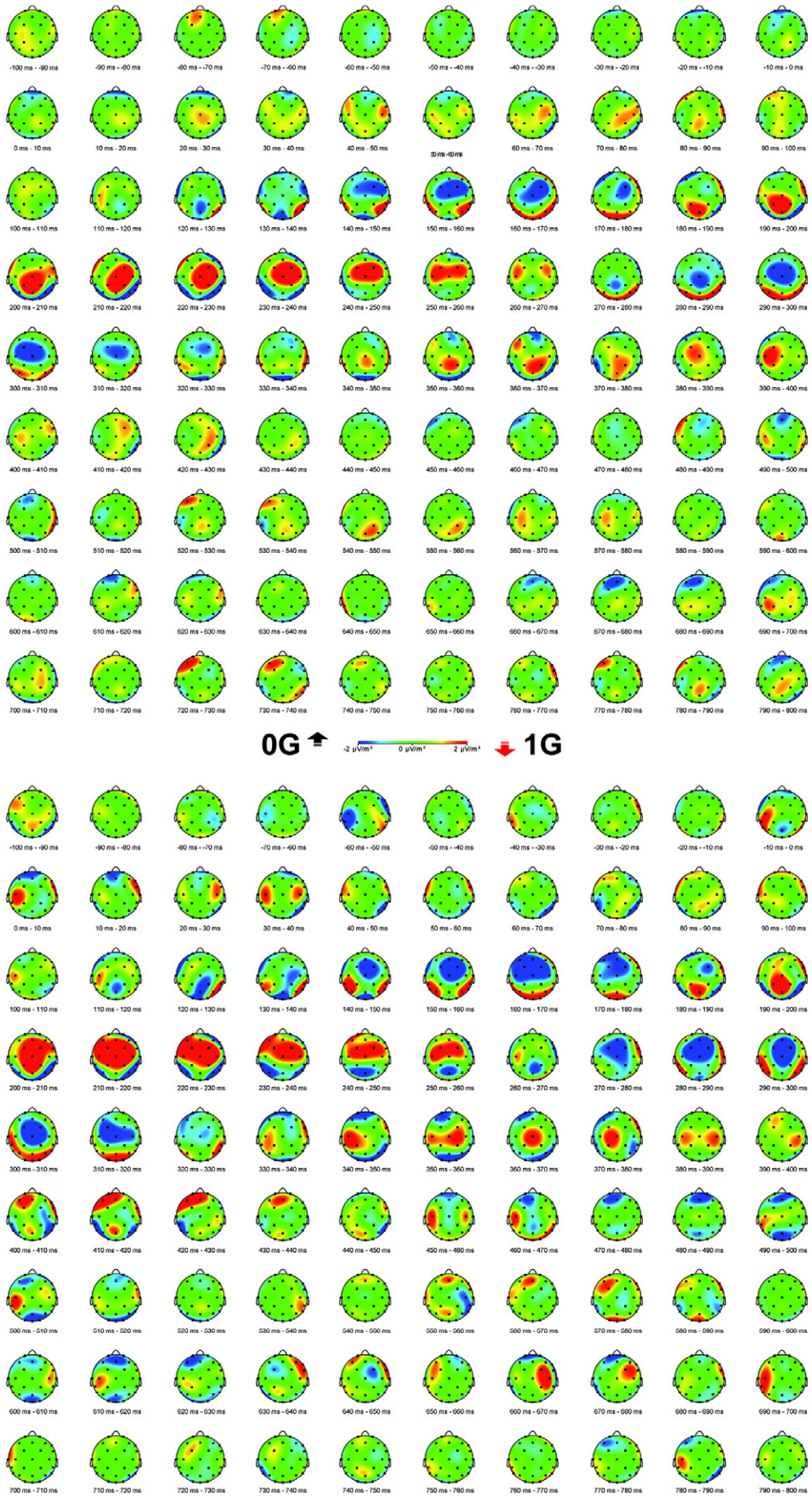

Fig. 3. Topographical map showing the changes in current source density (CSD) in steps of 10 ms over the scalp for the 0G condition (top) and the 1G condition (bottom). Clearly a negative peak at around 140–170 ms after stimulus onset (N1) in centro-frontal scalp areas as well as a positive peak (P2) across centro-parietal scalp areas starting at 180 ms post-stimulus and redistributing to centro-frontal areas until 250 ms post-stimulus can be identified.

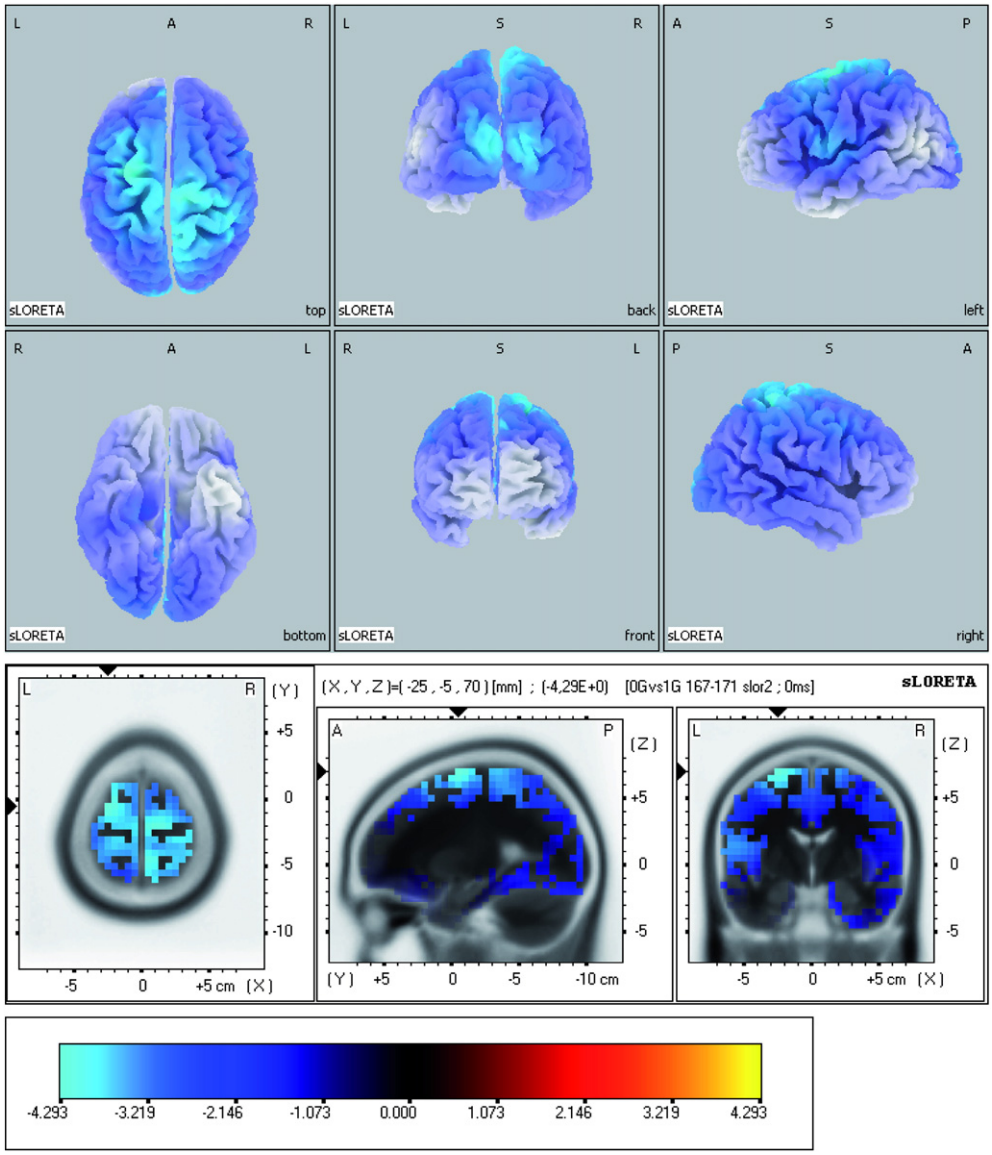

Fig. 4. Statistical parametric maps (SPM) of sLORETA differences in the ERP signal at 238 ms after stimulus onset (n = 39). Blue colors indicate decreased activity in the 0G condition. This was found to be localized in the superior frontal and middle frontal gyrus (BA 6).

3. Results

3.1. Reaction time

Results for group EASY showed no difference in reaction time between 0G and 1G ($F_{(1, 14)} = 2.63$, $p = .13$) nor between levels ($F_{(3, 42)} = .53$, $p = .66$). As expected, reaction time increased with increasing levels of difficulty in group DIFFICULT ($F_{(3, 78)} = 97.12$, $p < .001$), but interestingly this increase in reaction time was longer (slower) during the 1G condition ($F_{(1, 26)} = 18.90$, $p < .001$). Post-hoc analysis of the effects of gravity at the different levels ($F_{(3, 78)} = 5.58$, $p < .01$) in the DIFFICULT group revealed shorter reaction times in 0G compared to 1G at the more difficult levels (levels 3 and 4; Fig. 1).

3.2. Accuracy

A significant increase in the frequency of incorrect answers was observed in the 0G condition when combining both groups (main gravity effect: $F_{(1, 40)} = 4.70$, $p < .05$; Fig. 2). While there was a significant level * group interaction ($F_{(3, 120)} = 10.06$, $p < .001$), associated with an increased error rate in group DIFFICULT from level 1 to level 2 ($p < .001$) and level 2 to 3 ($p < .001$), no level * gravity * group interaction was observed ($F_{(3, 120)} = 1.08$, $p = .36$; Fig. 2).

3.3. ERP components/signal topography

As usual for visual stimuli [25], in both conditions (0G/1G) a clear N1 could be detected at around 140–170 ms after stimulus onset over centro-frontal scalp areas, followed by a P2 starting at around 180 ms at parieto-occipital location and shifting towards centro-frontal scalp areas until 250 ms after stimulus onset (Fig. 3).

T-statistics comparing ERPs of the 0G and 1G conditions in between 200 and 300 ms after stimulus onset using the sLORETA software revealed a significant difference ($p < .05$; $t = -4.15$; $t_{critical\ (two\text{-}tailed)} = 4.04$) for the time point 238 ms after stimulus onset. Comparing the calculated cortical distribution maps revealed a significantly lower activity

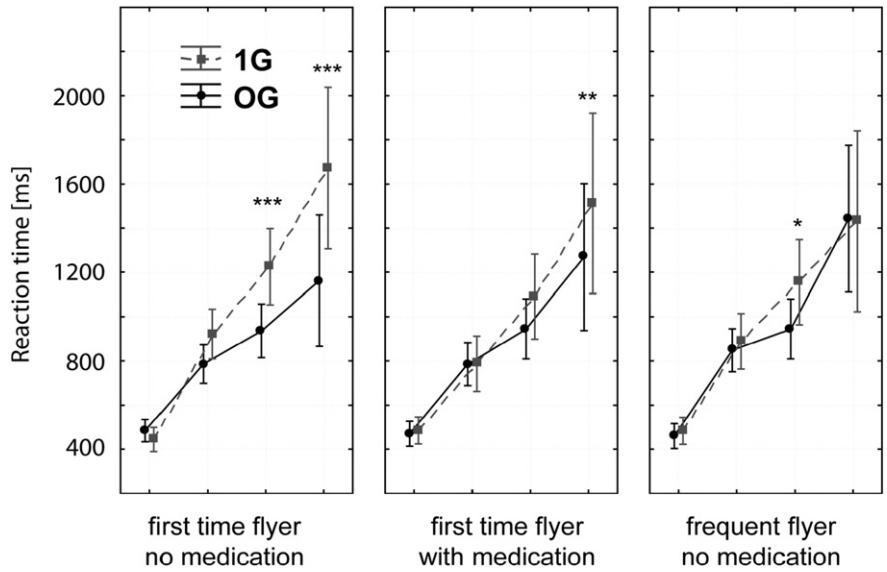

Fig. 5. Within each of the subgroups who performed the DIFFICULT protocol, reaction time performance tended to be improved at the higher levels of difficulty during the 0G condition (solid line). *p < .05, **p < .01, ***p < .001 between 0G and 1G.

in 22 adjacent voxels of the superior frontal and middle frontal gyrus (Brodman area 6) at this time point for the 0G condition (Fig. 4).

No significant differences were found for ERPs between 0 and 100 ms ($p = 0.37$; $t_{critical} = 4.06$) or 100 and 200 ms after stimulus onset ($p = 0.36$; $t_{critical} = 4.04$).

Displayed are six orthogonal views of the cortex (top) and a slice view of significant areas including density related color scale (bottom). Images depicting SPMs seen from different perspectives are based on voxel-by-voxel t-values of differences. Structural anatomy is shown in gray scale. (L left, R right, A anterior, P posterior).

3.4. Experience and medication

Further analysis within the DIFFICULT group of the possible effects of medication and/or experience showed no differences between experienced flyers (>100 parabolas) and first time flyers, with or without medication, for reaction time ($F_{(2, 23)} = .11$, $p = .89$) but confirmed the improved reaction performance at the higher difficulty levels during 0G ($F_{(6, 69)} = 2.40$, $p < .05$; Fig. 5).

A differentiation between experienced flyers and first time flyers with/without medication on P2 amplitude on position Cz revealed no effect of experience or medication (experienced flyers ($n = 12$): 6.06 μV +/− 2.72; no medication ($n = 14$): 8.08 μV +/− 5.57; medication ($n = 8$): 6.56 μV +/− 2.21; $F_{(2, 31)} = .67$, $p = .52$).

4. Discussion

This study aimed to determine the effect of short term microgravity on cognitive function and associated neurocognitive markers. Previous studies have demonstrated no, or a slightly negative impact, of microgravity on cognitive performance [1,2,26,27]. To further understand these behavioral changes, we aimed to assess the related neurophysiological markers. The present study showed an improvement in behavioral markers (RT) during 0G, as well as a positive impact of microgravity on neural processing quantified by ERPs.

It is well known that stress levels increase during parabolic flight, as indicated by alterations in brain cortical activity and cortisol levels, as well as negative effects on perceived physical and emotional state [28–30]. To further investigate the effect of gravity per se, we attempted to control for this flight induced stress by ensuring that assessments of reaction time and EEG activity were made during repetitive and consecutive periods of normal gravity and microgravity, with each being separated by no more than 30 s. It is unlikely that stress levels changed significantly between these two short periods. Furthermore, the microgravity induced changes were independent of the participants' previous parabolic flight experience, as consistent trends were seen in both first time flyers, who tend to have higher stress levels during flight [30], as well as more experienced frequent flyers. In addition, the use of anti-nausea medication, which also minimizes the stress associated with parabolic flights, had no influence on the finding that microgravity lead to an improvement in reaction time.

As can be seen by the participants' performance during the more advanced levels, the more difficult the tasks becomes, the greater the cognitive load that seems to be necessary, which is reflected by an increase in reaction time. Of particular importance here is the decrease in reaction time during the microgravity condition while completing the more complex problem solving tasks (levels three and four). Whereas no difference was observed between micro- and normal gravity during the initial "easy" levels, there was a significant decrease in reaction time during the completion of the more difficult levels in microgravity. Importantly, this decrease in reaction time is not reflected in a decrease of accuracy as there was no loss of accuracy during the 0G condition.

It is important to note that this improvement in RT during the more difficult levels is not reflected by any changes in the initial central processing. Although N1 seems to be a bit more blurred under 1G (Fig. 3), no significant effect could be obtained neither in N1 amplitude nor latency. A large number of previous studies focused on factors modulating the amplitude of the visual N1, suggesting that the N1 reflects a benefit of correctly allocated attentional resources and that it is a manifestation of an important sensory gating mechanism of attention [31]. The results obtained in this study let us assume that these processes were not affected by changed gravity conditions. A missing effect in N1 amplitude might be due to the fact that approximately 80% of the stimuli used in this study could be characterized as single choice reaction time tasks and previous results showed that amplitude effects on the N1 are absent during simple reaction time tasks, which only requires participants to make a rapid response to stimuli [32], A potential limitation of this study is that the 22 s of microgravity during each parabola only allowed for a maximum of nine assignments to be solved during difficulty levels 3 and 4 and therefore we are not able to differentiate the VEPs between the different difficulty levels.

The P2 has traditionally been studied in the context of perception, with specific emphasis on how stimulus evaluation takes place. Using different paradigms, the visual P2 amplitude is modulated by general visual features of stimuli as size, color and/or orientation of a presented task [17,33]. When investigated using a visual search paradigm, an increase in P2 amplitude has shown to be an indicator for attention, memory and response selection [34]. Studies showing that P2 seems to play a major role in selective attention and even relates to the derivation of an emotional percept [35] let us assume that P2, as a response, seems to be multidimensional and reflects a sensitivity to stimulus features. Moreover and most importantly for the present study, the N1–P2 complex has been shown to be influenced by stimulus salience. When presented in a multimodal way (visual *and* auditory) the N1–P2 complex is more expressed in comparison to the presentation in a single mode (visual *or* auditory) [36]. As the visual features of the presented task in this experiment did not differ between 0G and 1G, it might be argued that the change in P2 amplitude is either caused by changes in selective attention (participants might have been more focused in 0G) or, as it has been shown that P2 amplitude is dependent on the amount of sensory input [36], the missing influence of gravity might be responsible for the suppressed N1–P2 complex. Whether this is connected to the finding of a decreased reaction time cannot be fully answered with our data. One might argue that less sensory input allows for a better attentional focus and less discrimination between multisensory inputs (e.g. in 1G, besides the mental arithmetic processing, information on body position has to be integrated into preparing the tapping response). In particular, the involvement of BA6, which is an important area related to the planning of coordinated movements, raises the question of whether the described effect could be attributed to changes in (the processing of) motor control. Although any effects of learning or adaptation can be ruled out as those should be visible in both 0G and 1G conditions, one might assume impairment of motor control in microgravity as existent motor programs need to be adjusted, which would obviously impede on central resources. However, this would likely result in an increase not a decrease in reaction time. Moreover, motor control seems to be less vulnerable in this experiment as there were no fine motor skills needed with the tablet providing a relatively large tapping area of approximately 15 ∗ 20 cm.

Although previous experiments have shown that there is an accumulation of blood volume to the brain during microgravity [15], a systemic influence (e.g. changes in conduction) can be ruled out as such an effect should have been present also in N1 amplitude.

It is important to mention that neither previous experience with microgravity nor anti-nausea medication had any influence on the presented phenomenon. As there is a broad discussion about the effects scopolamine, used as anti-nausea drug, on human performance, especially sensory perception, this study clearly demonstrates that this effect can be neglected.

5. Conclusion and limitations

This study demonstrates that previous observations of a decrease in cognitive performance during conditions of microgravity may not be directly related to microgravity itself. In the present study where the confounding influence of stress was controlled, reaction time as well as the underlying early neural processes seem to be enhanced in microgravity. Whether this effect is limited to the initial sensation of weightlessness remains to be determined and further experiments under permanent microgravity would be needed.

With the data presented here, we are only able to identify an effect of gravity levels (0G and 1G) on VEPs. Given the low number of repeated stimuli within each level, a further differentiation between levels of difficulty is not possible but should be considered in future studies. Special attention should be paid to the fact that the improvement in reaction times is not mirrored in the temporal resolution of the ERPs but seems to be connected only with the magnitude of the response. While the aim here was to use a more complex task, future studies should include classical tasks like the oddball paradigm or the flanker task, for which ERPs have been clearly described.

Acknowledgments

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